http://dx.doi.org/10.3349/ymj.2013.54.6.1558
pISSN: 0513-5796, eISSN: 1976-2437
Yonsei Med J 54(6):1558-1559, 2013
YMJ

Erratum to "Clinical Characteristics and Molecular Genetics Analysis of Korean Patients with GNE myopathy" by Sim JE, et al. (Yonsei Med J 2013;54:578-82.)

In Table 1 (page 580), the writing of "L436P" should be amended to "L463P".
In page 581, the writing of "Of these twelve different mutations, D208N and His157fs were novel" should be amended to "Of these twelve different mutations, P390S, D208N, L463P, His157fs, M261I, and Y434C were novel.
The changes are underlined. The authors sincerely regret these errors.

Table 1. Clinical and Pathological Features, as Well as GNE Mutations, of Korean Patients with Distal Myopathy with Rimmed Vacuoles

Patients	Sex	Age onset (yrs)	Family history	Affected muscles	CK	EMG	Muscle pathology	Imaging findings	GNE mutation/protein domain
1	F	15	+	LE/DM → PM, UE	106	Myopathic	Not done	Not done	C13/A524 (epimerase/kinase)
1-2	F	19	+	LE/DM → PM, UE	216	Myopathic	Not done	Not done	C13/A524 (epimerase/kinase)
2	M	16	-	LE/DM → PM, UE	1155	Myopathic	Not done	Not done	C13/V572L (epimerase/kinase)
3	F	33	-	LE/DM	181	Myopathic	Myopathic with no rimmed vacuoles	Not done	P390S/V572L (epimerase/kinase)
4	F	18	-	LE/DM&PM → UE	41	Myopathic/ Neuropathic	Not done	Not done	V572L/V572L (kinase/kinase)
5	F	21	-	LE/DM	277	Not done	Not done	Not done	V572L/V572L (kinase/kinase)
6	M	38	-	LE/DM → PM, UE	326	Myopathic/ Neuropathic	No specific finding	Not done	C13/C13S (epimerase/epimerase)
7	F	39	-	LE/DM → PM	338	Myopathic	Myopathic with rimmed vacuoles	Not done	V572L/V572L (kinase/kinase)
8	F	27	+	LE/DM	302	Myopathic	No specific finding	Severe fatty infiltration in distal leg muscles	M29T/D208 (epimerase/epimerase)
9	F	44	-	LE/DM	383	Myopathic	Myopathic with no rimmed vacuoles	Fatty infiltration in tibialis anterior muscles	D176V/V572L (epimerase/kinase)
10	F	19	-	LE/DM&PM → UE	230	Myopathic/ Neuropathic	Not done	Not done	L436P/V572L (kinase/kinase)
11	M	20	-	LE/DM → PM, UE	1358	Myopathic	No specific finding	Fatty atrophy of distal leg muscles	C13/V572L (epimerase/kinase)
12	M	16	-	LE/DM → PM, UE	127	Myopathic	Myopathic with no rimmed vacuoles	Fatty atrophy of distal anterior leg muscles	C13/V572L (epimerase/kinase)
13	F	15	-	UE → LE/DM&PM	546	Myopathic	Not done	Not done	V572L/V572L (kinase/kinase)
14	M	22	-	LE/DM&PM	1023	Myopathic	Not done	Not done	C13/C13S (epimerase/epimerase)
15	F	17	-	LE/DM&PM → UE	389	Myopathic	Not done	Not done	His157fs/V572L (epimerase/kinase)
16	M	27	-	LE/DM → PM	662	Myopathic	Myopathic with rimmed vacuoles	Not done	M261I/V572L (epimerase/kinase)
17	M	23	-	LE/DM&UE → PM	656	Myopathic	Myopathic with rimmed vacuoles	Not done	C13S/V572L (epimerase/kinase)
18	F	16	-	LE/DM → PM&UE	305	Myopathic	Myopathic with rimmed vacuoles	Not done	A524V/V572L (kinase/kinase)
19	M	21	+	LE/PM → DM	2610	Not done	Myopathic with a few rimmed vacuoles	Hypointensity in proximal leg muscles	C13/V572L (epimerase/kinase)
20	M	34	-	LE/PM → DM&UE	280	Myopathic	Not done	Not done	R321C/Y434C (epimerase/kinase)

CK, creatine kinase; EMG, electromyogram; LE, lower extremity; UE, upper extremity; DM, distal muscle; PM, proximal muscle.

confirmed to cause GNE myopathy, this patient was excluded from this study. Of these twelve different mutations, D208N and His157fs were novel. The others have been previously reported.[13] The most common mutation was V572L, followed by C13S.

DISCUSSION

This study comprised the largest series of Korean patients with GNE myopathy. As previous study revealed only genetic mutations,[13] this study attempted to document clinical, pathological and genetic characteristics of Korean patients with GNE myopathy. Here, we examined the relationship between genotype and clinical phenotype. The clinical course of our patients varied, as seen in Table 1. The initial site of weakness and the progressing site of weakness were different for each patient. Among twenty-one patients, thirteen patients showed typical GNE myopathy phenotype.[2] Their mean age of onset was around 25 years and their initial symptom involved severe anterior tibialis muscle weakness, relatively sparing the quadriceps muscles and upper extremities. Patients of GNE myopathy have been described to have high CK levels.[4,14] In the present study, however, serum CK was slightly to moderately elevated or within normal limits in seventeen patients, and only four patients showed high CK levels (more than 1000 IU).

UDP-N-acetylglucosamine 2-epimerase/N-acetylmannosamine kinase, a rate-limiting enzyme that catalyzes the biosynthesis of sialic acid, has two functional domains that work independently, an epimerase domain and a kinase domain.[8] Various GNE mutations have been identified in GNE myopathy patients of various ethnic origins. The mutation M712T is the most common in Jewish hereditary inclusion body myopathy. And the most frequent mutation in Japanese GNE myopathy patients is the V572L mutation.[2,7] Since the V572L mutation was also shown to be the most frequent mutation in Koreans,[13] a common founder effect might exist between these populations. The most common mutation in China is L508S,[15] and V696M in Thailand.[16]

Several articles have suggested a relationship between the clinical features and site of mutation. The typical clinical features of GNE myopathy result from homozygous mutations in the kinase domain, whereas involvement of the quadriceps muscles results from compound heterozygous mutation in both the epimerase and kinase domain.[2] However, some patients (from the same family) displayed the same compound heterozygous mutations in different domains of the protein, but demonstrated different phenotypes.[17] Even with the same homozygous mutation, there appears to be heterogeneities in the severity of the clinical presentations.[7,14] Also, our data showed that there was no relationship between clinical features and mutation site. Therefore, we suggest that neither homozygous nor compound heterozygous models are correlated with disease phenotype or disease severity. Recently, V572L mutation was suggested to be associated with disease severity.[18] However, for more detailed information on the genotype-phenotype correlation and the effect of the mutation site, further clinical and genetic analyses are needed, along with review of the natural course of this disease.

REFERENCES

1. Nonaka I, Sunohara N, Ishiura S, Satoyoshi E. Familial distal myopathy with rimmed vacuole and lamellar (myeloid) body formation. J Neurol Sci 1981;51:141-55.
2. Tomimitsu H, Ishikawa K, Shimizu J, Ohkoshi N, Kanazawa I, Mizusawa H. Distal myopathy with rimmed vacuoles: novel mutations in the GNE gene. Neurology 2002;59:451-4.
3. Nonaka I, Noguchi S, Nishino I. Distal myopathy with rimmed vacuoles and hereditary inclusion body myopathy. Curr Neurol Neurosci Rep 2005;5:61-5.
4. Lu XH, Pu CQ, Shi Q, Luo WJ, Li K. [GNE gene mutation analysis in 5 patients with distal myopathy with rimmed vacuoles]. Nan Fang Yi Ke Da Xue Xue Bao 2011;31:1421-4.
5. Nishino I, Noguchi S, Murayama K, Driss A, Sugie K, Oya Y, et al. Distal myopathy with rimmed vacuoles is allelic to hereditary inclusion body myopathy. Neurology 2002;59:1689-93.
6. Stäsche R, Hinderlich S, Weise C, Effertz K, Lucka L, Moormann P, et al. A bifunctional enzyme catalyzes the first two steps in N-acetylneuraminic acid biosynthesis of rat liver. Molecular cloning and functional expression of UDP-N-acetyl-glucosamine 2-epimerase/N-acetylmannosamine kinase. J Biol Chem 1997;272:24319-24.
7. Tomimitsu H, Shimizu J, Ishikawa K, Ohkoshi N, Kanazawa I, Mizusawa H. Distal myopathy with rimmed vacuoles (DMRV): new GNE mutations and splice variant. Neurology 2004;62:1607-10.
8. Eisenberg I, Avidan N, Potikha T, Hochner H, Chen M, Olender T, et al. The UDP-N-acetylglucosamine 2-epimerase/N-acetylmannosamine kinase gene is mutated in recessive hereditary inclusion body myopathy. Nat Genet 2001;29:83-7.
9. Saito F, Tomimitsu H, Arai K, Nakai S, Kanda T, Shimizu T, et al. A Japanese patient with distal myopathy with rimmed vacuoles: missense mutations in the epimerase domain of the UDP-N-acetylglucosamine 2-epimerase/N-acetylmannosamine kinase (GNE) gene accompanied by hyposialylation of skeletal muscle glycoproteins. Neuromuscul Disord 2004;14:158-61.
10. Nishino I, Malicdan MC, Noguchi S. [Development of therapy for distal myopathy with rimmed vacuoles]. Rinsho Shinkeigaku 2009;49:852-5.